# Effects of Acute Cardiovascular Exercise on Memory

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# Introduction

**Interest**

Effects of various jogging durations on memory game completion length

Combining group interests of cognitive science and physical activity

**Expectations**

Both durations of exercise could increase memory performance, but shorter exercise durations may have better improvements due to less body fatigue

# Design Of The Experiment

We decided that we wanted our experiment to be a randomized complete block design (RCBD). In order to do this, we forwent pairwise comparisons and just used the difference in the islander's scores in the memory game from before and after they completed their assigned treatment.

- Two treatments
  - 5 minutes of light jogging,
  - 30 minutes of light jogging
- Three blocks done by island

# Random Sampling Procedure

1. Use a random number generator in R in order to randomly select one city from each island
2. Gather the names of people in each town who were in our desired age range of 30 - 35 (using birthdays) from each town's registry (to mitigate the potential nuisance factor)
3. Randomly select 36 names from each towns registry for a total n of 108
4. Assign half of the selected population from each town to the 30 minutes condition and the other half to the 5 minute condition

# Limitations With Sampling

We had to eliminate ghosts and those who did not consent from our list of viable persons, then generate a new set of random numbers and reselect the individuals. Additionally, the towns registry only keeps track of the births on the island so it is possible that some islanders moved to a different island after birth, which could interfere with our blocking

# Power And Sample Size Calculations

We initially wanted the power level of our experiment to be 0.9. Once we ran the experiment, we found MSE = 61.55 and chose our d as 3.25 seconds for the maximum average mean distance. From this, we got an effect size of 0.41. Using the pwr.anova.test, we determined that the best sample size would be 32 responses per island for a total of 96 participants. Since our original n was 36, this confirmed that our sample size was appropriate for the experiment and that we had an acceptable level of power in our results.

```{r}
library(pwr)
pwr.anova.test(k = 3, n = 36, p = 0.8)
```

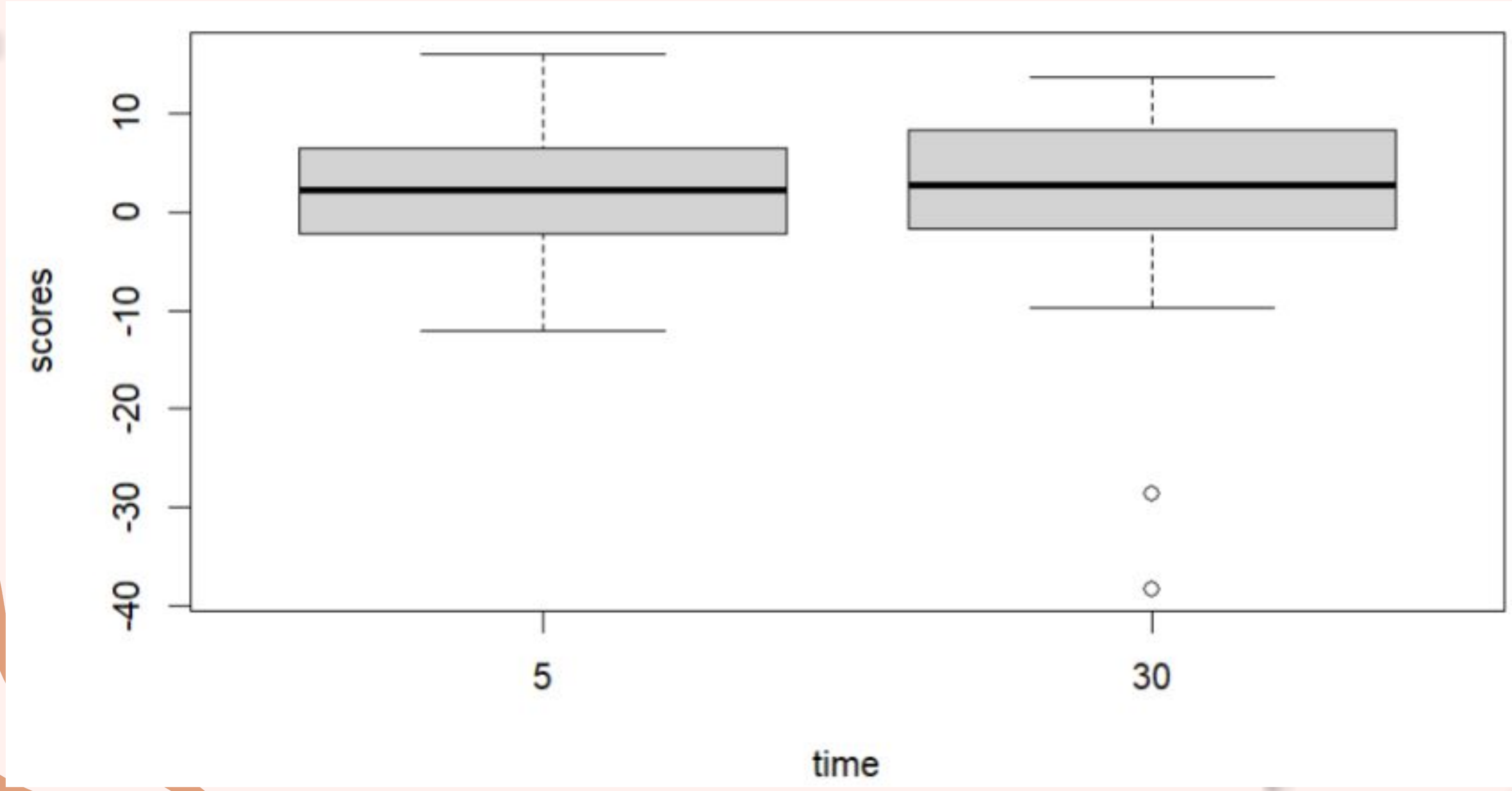
```
     Balanced one-way analysis of variance power calculation

              k = 3
              n = 36
              f = 0.303013
      sig.level = 0.05
          power = 0.8

NOTE: n is number in each group
```

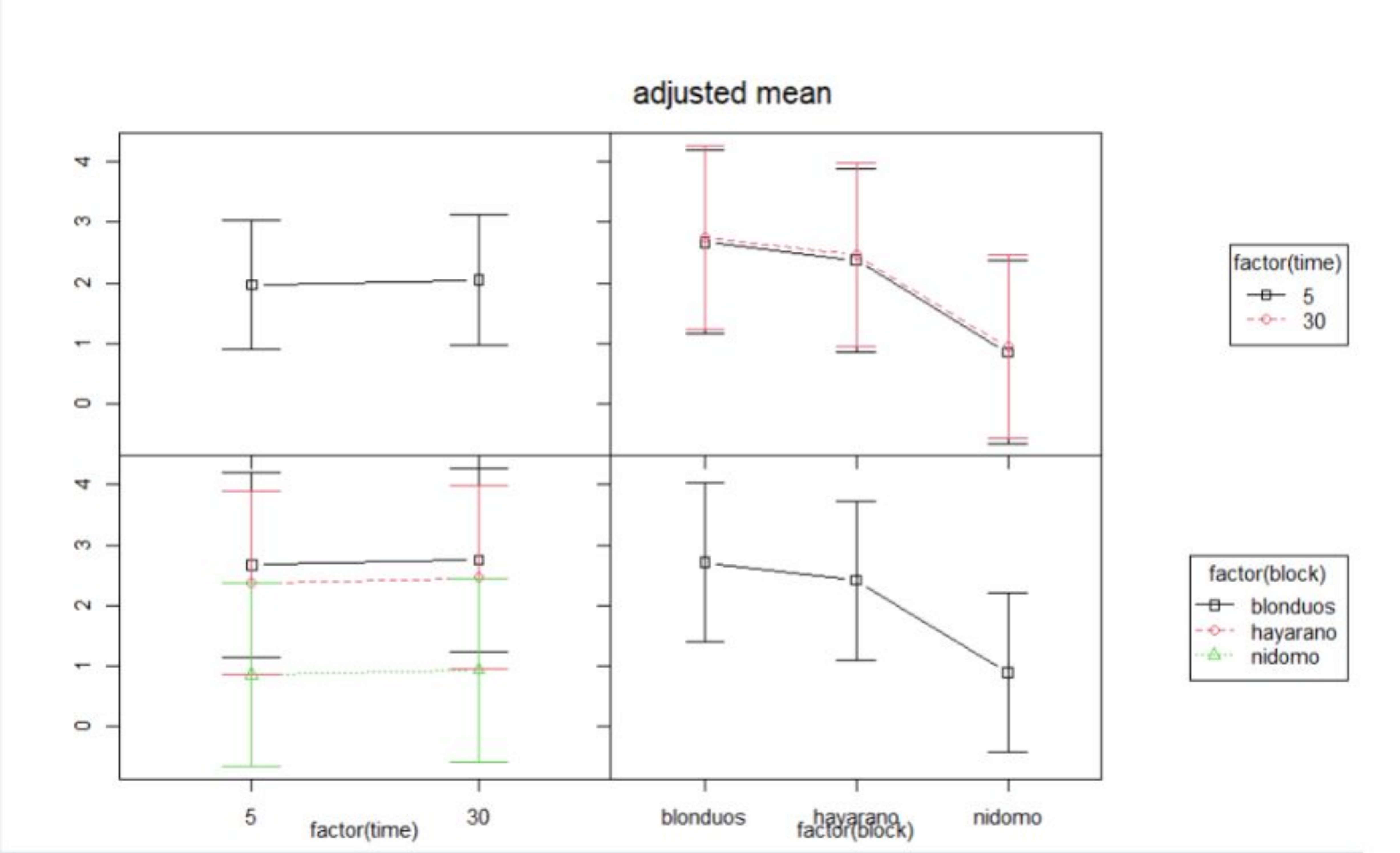
# Results and Interpretation–ANOVA

| | Df | Sum Sq | Mean Sq | F value | Pr(>F) |
|---|---|---|---|---|---|
| factor(time) | 1 | 0 | 0.20 | 0.003 | 0.954 |
| factor(block) | 2 | 68 | 34.22 | 0.551 | 0.578 |
| Residuals | 104 | 6456 | 62.08 | | |



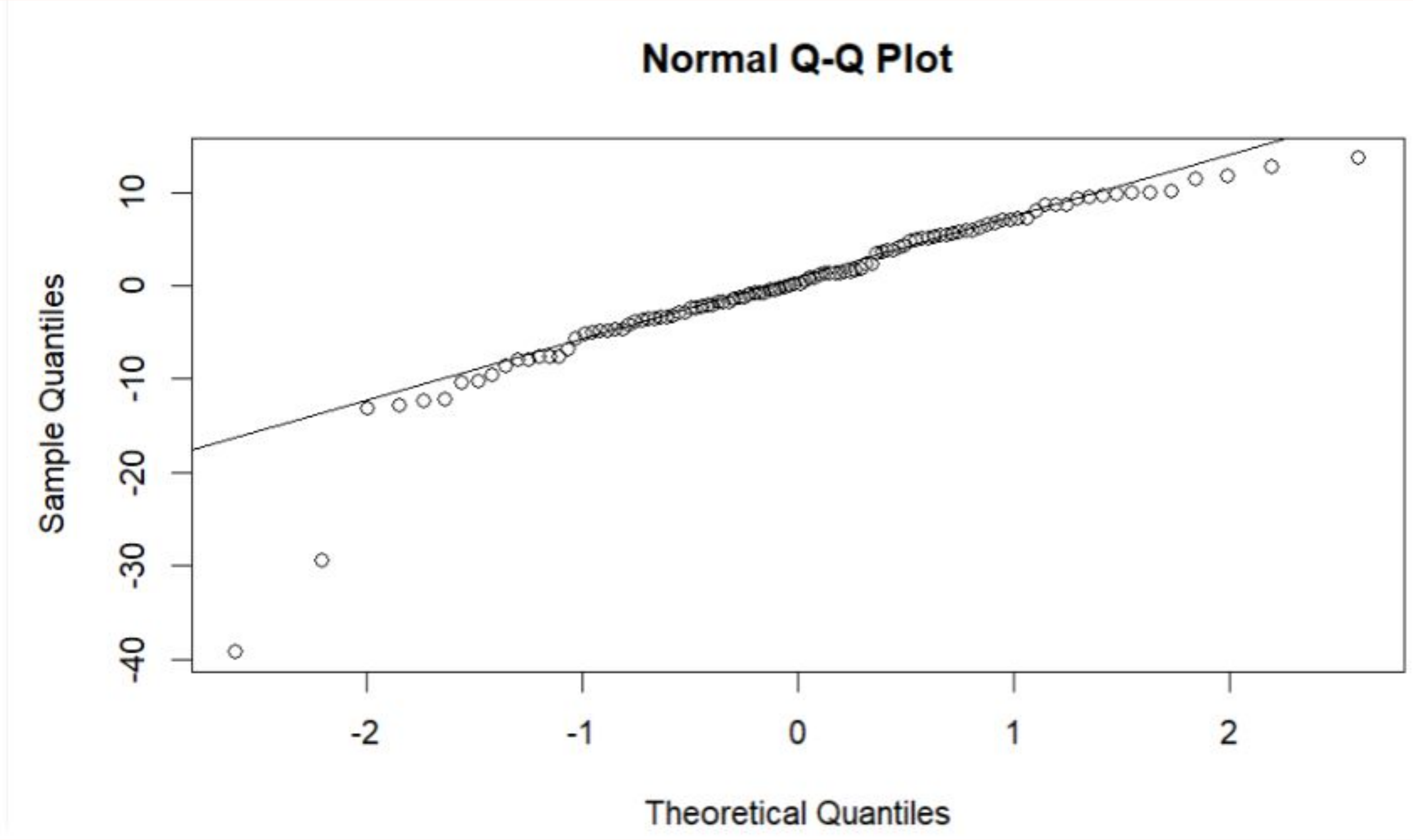

- Both time of jogging and block (island) seem to be statistically insignificant
- Low SSE: Block is unnecessary
- This means performing light jogging for 30 minutes or 5 minutes before taking a memory test will not have any effect on the scores of the memory game.

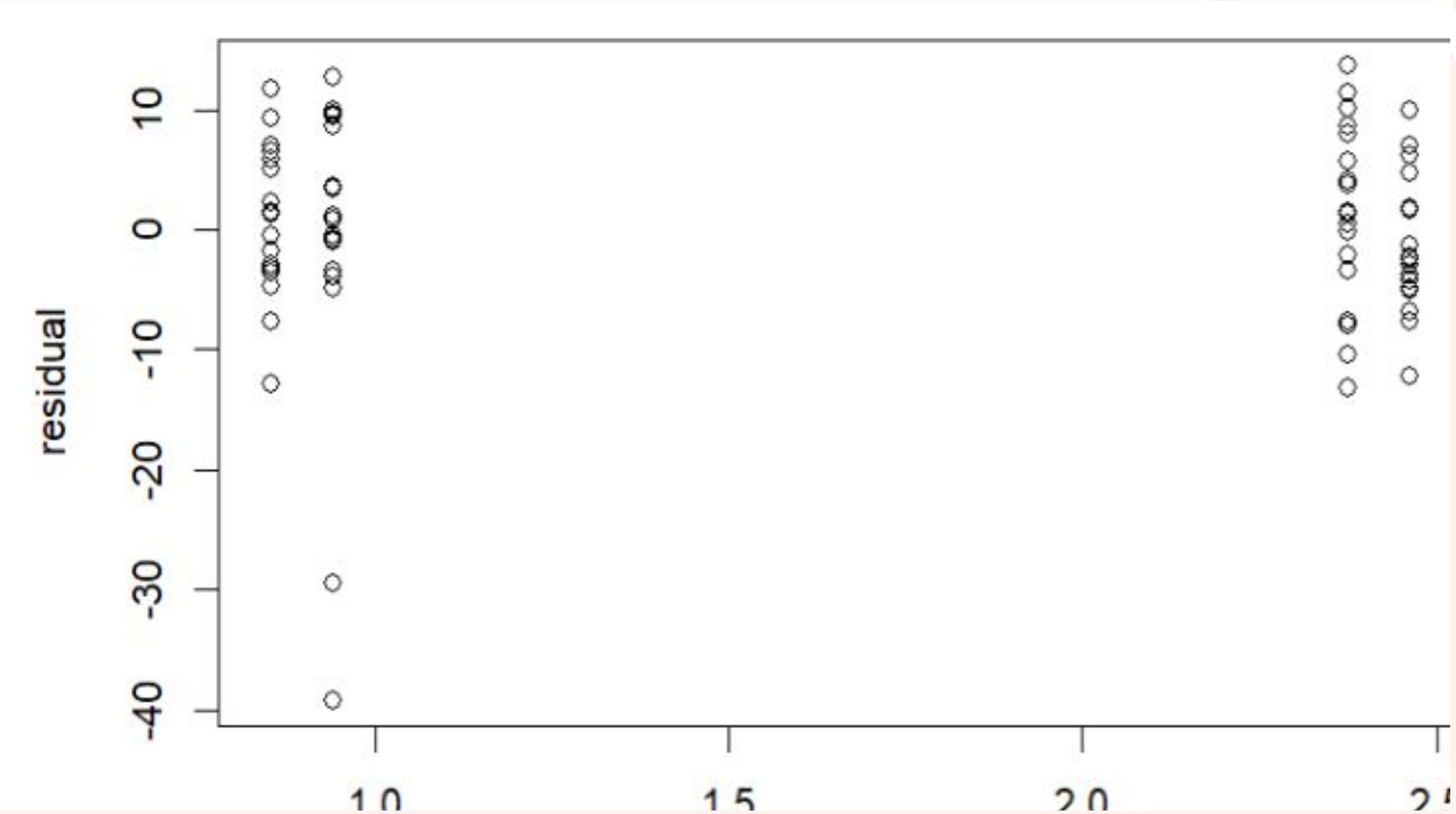
# Results and Interpretation–Interaction Plots




- Parallel lines–no interaction
- Nidomo seems to have the lowest scores in the memory game, but this difference is insignificant in the scale of the whole experiment

# Results and Interpretation - Model Adequacy




- ☀️ No specific pattern appears
- ☀️ Normality and equal-variance assumptions are satisfied

# Discussion

**Conclusion**

- Neither the island a participant was born on, nor the amount of time they spent jogging had a significant impact on their scores in the memory game
- Normality and constant variance are satisfied, meaning the model is valid

**Limitations**

- Randomization in the sampling process — forced to replace any invalid participants by conducting another random number generation
- Unable to control for nuisance factors which may have been impacting memory (e.g. lifestyle habits, genetic components, climate)

# Outside Literature

Fitness tracking reveals task-specific associations between memory, mental health, and physical activity - cardiovascular exercise improves recall on naturalistic memory tasks

Acute and Chronic Exercise Effects on Human Memory: What We Know and Where to Go from Here - shows a significant impact by both acute (or a short duration) and chronic (a long duration) exercise on episodic memory

The Effects of Acute Cardiovascular Exercise on Memory . . . - several weeks or months of exercise results in an increase in performance, no beneficial effect of a single cardiovascular exercise on memory consolidation

Association between habitual physical activity on episodic memory strategy use and memory controllability - no association between physical activity and memory strategy use

Overall, this experiment addresses only a small complexity of this overarching topic, and our results prove to be consistent with studies which isolate the impacts of acute cardiovascular exercise on memory consolidation.

# References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